Exercise class 6

1. *Direction fields*. Sketch the direction field (slope field) of the first order ordinary differential equation

$$y'(x) = x + y(x)\,.$$

Draw the isoclines and defined by $f(x,y) = x + y = C$ for a few values of $C = 0, 1,$ $-1, \dots$ together with the direction vectors directions along them. Sketch a few $y(x)$ contours.

2. *Simple differential equations.* Kreyszig 1.1, Problems 1, 2, 3, 6 (not first order!).
3. *Simple initial value problems.* Kreyszig 1.1, Problems 10, 12, (14), 18.
4. *Radioactive decay, half-life. Kreyszig 1.1, Problem 18.*
5. *Dropped ball. The differential equation describing a falling body with air resistance* reads (with the positive direction pointing downwards)

$$\dot{v}(t) = g - k\,v(t)^2\,.$$

a) Draw and analyse the direction field of this equation.

b) Verify that the function

$$v(t) = \bar{v}\,\frac{1 - C\,e^{-2k\bar{v}\,t}}{1 + C\,e^{-2k\bar{v}\,t}}$$

solves this equation with $\bar{v} = \sqrt{g/k}$ being the terminal velocity.

c) Show that C is fixed by the initial velocity $v(0) = v_0 > 0$,

$$C = \frac{\bar{v} - v_0}{\bar{v} + v_0}\,.$$

d) Show that for large times, $t \gg k\bar{v},\ v(t) \approx \bar{v}\,(1 - 2Ce^{-2k\bar{v}\,t})$.

e) Show that for short times, $t \ll k\bar{v}, v(t) \approx v_0 + gt - kv_0^2 t$.

f) Assume that $v_0 \gg \bar{v}$. Then g can be neglected for short times. Show that the solution of the resulting simpler equation is

$$v(t) = \frac{v_0}{1 + kv_0 t}\,.$$

Note that the short time behaviour of this function agrees with that found in part e) with $g = 0$.